ISAIAH KOL

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EDUCATION

University of Vermont | Burlington, VT — Aug 2024 – Present

Bachelor of Science, Mechanical Engineering | **Expected May 2028**

- GPA: 3.84 | 3x Dean's List | Minor: Pure Mathematics

EXPERIENCE

CNC Machinist Intern | **Preci Manufacturing** | Winooski, VT — May 2026 – Aug 2026

- Rotated through turning, 5-axis milling, and quality control in an aerospace manufacturing environment.
- Executed CNC setups, assembling tooling and workholding, and adjusting geometry for first-off inspection.
- Inspected parts and adjusted tool offsets during production to maintain tolerances as tight as ± 0.0001".
- Supported implementation of automation, verifying workholding, part handling, and process repeatability.
- Performed QC inspections using GD&T and metrology equipment including CMMs, optical comparators, indicators and precision gaging; verified material/process certifications and traceability.

UVM FabLab | Burlington, VT

Operations Manager — Dec 2025 – Present

- Lead lab operations, balancing hands-on fabrication with technical direction and final decision making.
- Manage preventative maintenance and SOP development across all lab equipment.
- Lead biweekly staff meetings to review lab performance, address issues, and implement changes.
- Troubleshoot and repair 3D printers and laser cutters, resolving issues escalated by technicians.

Technician — Sep 2025 – Dec 2025

- Contributed to 600+ fabrication tickets in one semester, operating FDM/SLA printers and laser cutters.
- Advised students and researchers on CAD, DFM, material selection, and fabrication strategies.

Gap-Year: VEX Robotics Team Coach | Thetford, VT — Aug 2023 – Jun 2024

- Mentored students aged 12-18 in CAD, design process, safe fabrication, and competition strategy.

VEX Robotics Team 4886-B | Various Locations, USA — Sep 2017 – Jun 2023

- Shared responsibility for robot design, fabrication, controls, and documentation on a two-person team.
- Four-time VEX World Championship competitor; 2023 NH/VT State Champion; recipient of three Design Awards and three Excellence Awards at the regional (NH/VT) level.

STEM Intern | Hypertherm & Fujifilm Dimatix | Lebanon, NH — Aug 2021 – Jan 2022

- Developed foundational CAD, manual machining, and manufacturing skills through engineering projects.

PROJECTS

6-DOF Robot Arm — Mar 2026 – Present

- Designed and built a robot arm integrating custom cycloidal reducers, belt drives, and stepper motors.
- Programmed multi-axis motion control using Arduino mega, digital stepper drivers, and a 24V PSU.
- Built and tested two complete iterations, upgrading control hardware and redesigning mechanical assemblies in SOLIDWORKS to improve load capacity, weight distribution, and packaging.

Cycloidal 21:1 Reduction Gearbox — Jan 2026 – Present

- Designed and 3D printed a cycloidal reducer for NEMA 17 stepper motors, used throughout the 6-DOF arm.
- Iterated cycloidal geometry, material selection, and FDM print tolerances to minimize backlash.
- Integrated dual cycloidal discs, output pins, bearings, and an eccentric shaft into a 20mm axial package.

SKILLS

Design: SOLIDWORKS, AutoCAD, DFM, GD&T, Technical Drawings and Documentation

Manufacturing: CNC & Manual Machining, FDM/SLA 3D Printing, Laser Cutting, Metrology

Software: Arduino, Python, MATLAB, FDM Slicing Software, Excel, MS Office Suite